

City Segmentation Using Deep Learning

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1. Project Background

Urbanization has increased rapidly in recent decades, creating a growing demand for efficient and accurate methods of monitoring urban areas. City segmentation is an important task in Geographic Information Systems (GIS) and remote sensing that enables the identification and classification of urban regions from satellite imagery. Accurate city segmentation supports applications such as urban planning, infrastructure development, environmental monitoring, disaster management, and land-use analysis.

Traditional approaches for analyzing satellite imagery rely heavily on manual interpretation, which is time-consuming, labor-intensive, and difficult to apply to large geographic areas. Recent advances in deep learning have significantly improved semantic segmentation by enabling pixel-level classification of satellite images. Modern deep learning models can automatically learn spatial features from high-resolution satellite imagery and generate accurate segmentation maps by classifying urban land-cover categories such as buildings, roads, vegetation, water bodies, barren land, forests, and agricultural areas.

Among the state-of-the-art semantic segmentation models, SegFormer, Segment Anything Model (SAM), and DeepLabV3 have demonstrated promising performance in image segmentation tasks. This project aims to implement and compare these three models using the LoveDA Urban dataset to accurately segment urban land-cover classes, including buildings, roads, vegetation, water bodies, barren land, forests, and agricultural areas, and determine the most effective approach for city segmentation.

2. Related Work

Semantic segmentation has become one of the most important techniques in remote sensing for extracting urban features from satellite imagery. Recent advances in deep learning have significantly improved segmentation accuracy for applications such as urban planning, land-use mapping, and environmental monitoring.

Fetene et al. (2026) conducted a systematic literature review on semantic segmentation for smart city development and reported that models such as U-Net and DeepLabV3+ achieved excellent performance on satellite imagery. The study also identified transformer-based architectures as promising future approaches.

Song (2023) demonstrated that deep learning models can accurately segment urban regions from heterogeneous aerial imagery, producing reliable results for GIS applications. Similarly, a study published in PLOS ONE (2024) showed that deep learning-based semantic segmentation provides accurate pixel-level classification of urban scenes and improves city mapping.

Motivated by these studies, this project compares SegFormer, Segment Anything Model (SAM), and DeepLabV3 using the LoveDA Urban dataset to identify the most effective model for city segmentation.

3. Objectives

- To develop deep learning models for automatic city segmentation using satellite imagery.
- To implement and compare SegFormer, SAM, and DeepLabV3 for city segmentation.
- To generate pixel-level segmentation maps of urban areas.
- To evaluate the performance of the models using semantic segmentation metrics.
- To identify the most suitable deep learning model for GIS-based city segmentation.

4. Dataset

Table 1 presents the dataset used in this project.

Table 1: Dataset used for city segmentation

Dataset	Description	Classes	Source
LoveDA (Urban Subset)	LoveDA (LoveDA: A Remote Sensing Land-cover Dataset for Domain Adaptive Semantic Segmentation) is a publicly available high-resolution remote sensing dataset developed for semantic segmentation research. It contains both urban and rural satellite images with pixel-level annotations for seven land-cover classes. This project uses only the Urban subset, which contains high-resolution urban satellite images and corresponding ground-truth segmentation masks for training and evaluating deep learning models.	Background, Building, Road, Water, Barren, Forest, Agricultural	https://github.com/Junjue-Wang/LoveDA

5. Methodology

System Overview and Workflow

Figure 1 illustrates the overall workflow followed in this project, from dataset collection through preprocessing, model implementation, training, evaluation, and result interpretation.

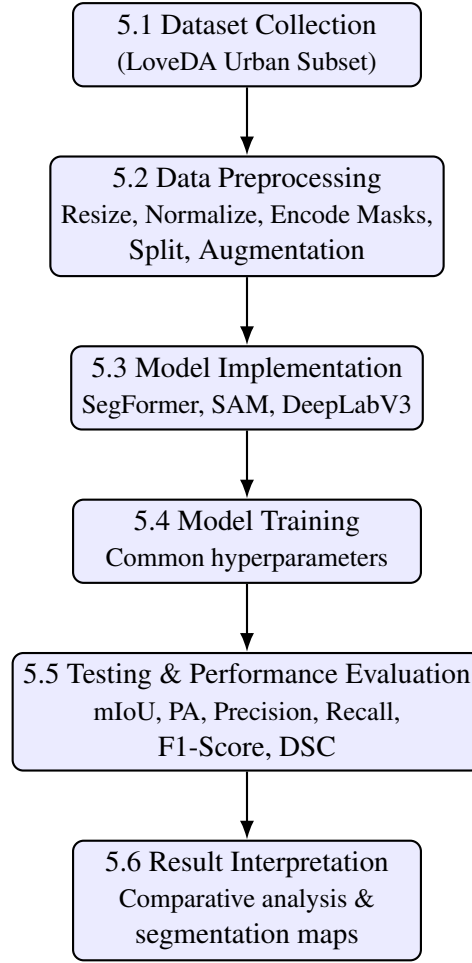


Figure 1: Workflow of the proposed city segmentation methodology

5.1 Dataset Collection

The LoveDA dataset will be downloaded from its official repository. Since the focus of this project is city segmentation, only the Urban subset of the dataset will be used. The dataset contains satellite images along with pixel-level annotated segmentation masks.

5.2 Data Preprocessing

The collected images will undergo preprocessing before training the models.

- Resize images to a fixed input resolution.
- Normalize pixel values.
- Verify and encode segmentation masks.
- Split the dataset into training, validation, and testing sets.
- Apply data augmentation techniques including:
 - Random horizontal and vertical flipping
 - Rotation

- Random cropping
- Scaling

These preprocessing techniques improve model robustness and reduce overfitting.

5.3 Deep Learning Model Implementation

Three semantic segmentation models will be implemented and compared, as summarized in Table 2.

Table 2: Deep learning models used for city segmentation

Model	Description
SegFormer	A transformer-based semantic segmentation model that combines hierarchical transformer encoders with a lightweight decoder. It efficiently captures both local and global contextual information, making it suitable for high-resolution satellite imagery.
Segment Anything Model (SAM)	A foundation segmentation model developed by Meta AI that generates high-quality segmentation masks using image prompts. It will be evaluated for its ability to segment urban regions from satellite imagery.
DeepLabV3	A CNN-based semantic segmentation model that employs Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information. It has demonstrated strong performance in remote sensing image segmentation.

5.4 Model Training

Each model will be trained using the same training dataset under similar experimental settings. Hyperparameters such as learning rate, batch size, optimizer, and number of epochs will be selected to ensure a fair comparison among the models.

5.5 Testing and Performance Evaluation

The trained models will be evaluated using the following semantic segmentation metrics:

- Mean Intersection over Union (mIoU)
- Pixel Accuracy (PA)
- Precision, Recall, and F1-Score
- Dice Similarity Coefficient (DSC)

These metrics will be used to compare the performance of SegFormer, SAM, and DeepLabV3.

5.6 Result Interpretation

The trained models will generate pixel-level segmentation maps of urban satellite images. The outputs produced by SegFormer, SAM, and DeepLabV3 will be analyzed both visually and quantitatively. The evaluation metrics will be used to compare segmentation accuracy and identify the most effective model for city segmentation. The results will demonstrate the applicability of deep learning techniques for GIS-based urban mapping.

6. Expected Outcomes

- A preprocessed LoveDA Urban dataset for model training.
- Implementation of SegFormer, SAM, and DeepLabV3 for city segmentation.
- Pixel-level segmentation maps of urban satellite imagery.
- Comparative analysis of the three deep learning models.
- Identification of the most effective model for city segmentation.
- GIS-based visualization and interpretation of segmented urban areas.

7. Conclusion

City segmentation from satellite imagery plays a vital role in supporting urban planning, infrastructure development, environmental monitoring, and disaster management. Manual interpretation of satellite imagery is slow and difficult to scale, which highlights the need for automated, deep learning-based approaches capable of accurately classifying urban land-cover at the pixel level.

This project proposes the implementation and comparative evaluation of three state-of-the-art semantic segmentation models, namely SegFormer, the Segment Anything Model (SAM), and DeepLabV3, using the Urban subset of the LoveDA dataset. By training and testing these models under consistent experimental conditions and evaluating them using standard metrics such as mIoU, Pixel Accuracy, Precision, Recall, F1-Score, and the Dice Similarity Coefficient, the project aims to identify the model that offers the most accurate and reliable performance for city segmentation.

The outcomes of this project, including the preprocessed dataset, trained models, pixel-level segmentation maps, and comparative analysis, are expected to demonstrate the practical applicability of deep learning techniques in GIS-based urban mapping. The findings will provide useful insights for selecting appropriate deep learning architectures for future remote sensing and urban analysis applications.

8. References

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